Disclaimer/credit. These questions have largely been ~~stolen~~ borrowed from natural sciences maths and numerical methods courses.

1) For the function $f(x)$, consider Taylor expansions about $f(x+\Delta x)$ and $f(x+2\Delta x)$ to derive:

   (a) A first order approximation to the derivative of f at x.

   (b) A second order approximation to the derivative of f at x.

2) (a) Write down the Taylor series for a function f evaluated at $x+h$ in terms of $f(x)$ and its derivatives evaluated at x. Use this result to show that if x_0 is an approximate solution of the equation $f(x) = 0$, then a better approximation is given, in general, by $x = x_1$ where

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

   (b) Sketch the graph of $g(x) = x^3 - 3x^2 + 2$.

   Use the above formula with an initial guess $x_0 = 2.5$ to obtain an improved estimate, x_1 of the highest root of the equation $g(x) = 0$. Apply the method a second time to get a further estimate x_2 (you may leave your answer in rational form).

   Give a sketch showing the progress of the iterations and demonstrate that the sequence $x_0, x_1, x_2, x_3, \ldots$ will converge to the root.

   (c) To which root (if any) do you think the Newton–Raphson method will converge if:

   (i) $x_0 = 1.5$

   (ii) $x_0 = 1.9$

   (iii) $x_0 = 2.0$

   Justify your answers with sketches.

3) (a) Show that $f(x) = x^4 + x - 1$ has a real root α in the interval $[0.5, 1.0]$.

   (b) Starting with the interval $[0.5, 1.0]$, use interval bisection twice to find an interval of width 0.125 which contains α.

   (c) Taking 0.75 as a first approximation, apply the Newton–Raphson process twice to $f(x)$ to obtain an approximate value of α. Give your answer to 3 decimal places.

4) Given $h > 0$, let Euler's method be applied to calculate the estimates $\{y_n\}$ of $y(nh)$ for each of the differential equations

$$y' = -\frac{y}{1+t} \quad \text{and} \quad y' = \frac{2y}{1+t}, \qquad 0 \le t \le 1,$$

   starting with $y_0 = y(0) = 1$ in both cases. By using induction and by cancelling as many terms as possible in the resultant products, deduce simple explicit expressions for y_n, which should be free from summations and products of n terms. Hence deduce the exact solutions of the equations from the limit $h \to 0$ and $nh \to t$. Verify that, for $nh \in [0, 1]$, the magnitude of the errors $y_n - y(nh)$ is at most $\mathcal{O}(h)$.

5) Assuming that $\mathbf{f}$ satisfies the Lipschitz condition and possesses a bounded third derivative in $[0, T]$, apply the method of analysis of the Euler method, given in the lectures, to prove that the trapezoidal rule

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \tfrac{1}{2}h\big[\mathbf{f}(t_n, \mathbf{y}_n) + \mathbf{f}(t_{n+1}, \mathbf{y}_{n+1})\big]$$

   converges and that $\|\mathbf{y}_n - \mathbf{y}(t_n)\| \le ch^2$ for some $c > 0$ and all n such that $0 \le nh \le T$.